CIPHER: Contextual Instruction Precedence and Hierarchy Evaluation for Repositories

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***Abstract*—Repository instruction files such as AGENTS.md guide AI coding agents, and a nested file may legitimately refine or replace guidance from its ancestors, so an apparent disagreement is not necessarily a defect. CIPHER defines a task-conditioned classification of instruction pairs under a declared directory-precedence profile into five labels: unresolved conflict, valid override, disjoint applicability, compatible, and insufficient evidence. From a public configuration dataset we built a traceable pilot set of 27 real instruction pairs from eight repositories at pinned commits, each with a context-changing variant and an order control, and a blinded annotation protocol. Human reference labels have not yet been collected, so this short paper reports a preliminary analysis, frozen by hash before predictions were opened, that needs none: accuracy on the 21 originals and variants whose label the profile fixes from file paths, and label-free measures elsewhere. With one local open-weight model, a direct prompt given the full evidence and a structured pipeline that computes applicability and precedence explicitly were both correct on all 21 profile-determined cases, whereas a keyword heuristic flagged 2 of 26 such cases as conflicts. No model judged any original or variant to be an unresolved conflict; the methods disagreed mainly on condition-based applicability, and swapping only the order of the two instructions changed 4 of 27 predictions for each model arm. The structured pipeline used more output tokens than the direct prompt (3589 versus 2177 per case) without a measurable gain. Human labels for the semantic cases are the next step.**

***Index Terms*—AI coding agents, AGENTS.md, instruction conflicts, repository mining, scope and precedence**

I. Introduction

AI coding agents read project guidance from natural-language files in the repository. The AGENTS.md convention allows one file per package and states that "the closest AGENTS.md to the edited file wins" [1]. Apparent disagreement between such files is therefore not always a defect. Suppose the root file says "use exactly four spaces per indentation level" and a file in `web/` says "use exactly two". For a new line in the root package the two rules may genuinely conflict; for `web/src/button.ts` the nested rule is a documented local override; and if the second rule lived in `api/`, it would not apply to that target at all. The wording is identical in all three situations; only the task's location and the files' placement differ.

A conflict detector can fail in both directions: it can miss incompatible obligations, or it can report a legitimate local exception as a defect and suggest deleting useful guidance. Existing detectors flag contradictions from text alone. In a study of 100 repositories, a model-based smell detector reported 28 conflicting-instruction cases, of which manual analysis by that study's first author confirmed 16 [2]; the keyword heuristics of Ruleherd [3] compare files without regard to scope. Whether scope or precedence explains such false alarms has not been tested.

CIPHER asks whether making applicability and precedence explicit improves the classification of apparent conflicts: how reliably humans separate the labels (RQ1), whether a structured pipeline beats a direct model given the same evidence (RQ2), and whether predictions respond correctly to relevant context changes and stay stable under irrelevant ones (RQ3). We contribute (1) a task-conditioned, five-label definition of the problem under a declared AGENTS.md loading profile; (2) a traceable pilot set of 27 real instruction pairs from eight repositories at pinned commits, each with one context-changing variant and one order control, and a blinded annotation protocol and tool for RQ1; and (3) a comparison of a keyword heuristic, a direct model given the full evidence, and a structured pipeline using the same local model. Human labels have not yet been collected, so Section VI reports a preliminary analysis without human labels: on cases whose label the profile fixes from directory structure, a direct model given the full evidence was already correct, so the structured pipeline added cost without measurable benefit there, while the methods disagreed on condition-based applicability. The protocol, data, prompts, raw outputs, and analysis code are released at https://github.com/4waiz/cipher.

II. Related Work

Agent context files have been characterized at scale [4], [5], and their effect on task success is contested [6]. Configuration Smells [2] treats conflicting instructions as a smell detected within one file; Context Rot [7] studies staleness against code. WIRE [8] grounds rules in spans and models activation conditions and decision surfaces, but reports no matched compatible-pair or priority-insertion controls. ContextCov [9] extracts scoped constraints and enforces them, and OctoBench [10] measures which source an agent follows under explicit conflicts. Guidefold [11] already resolves guidance with a most-specific-scope lattice and structural replays, while noting that human-labeled conflicts were still missing. A

factorial study found no detectable effect of injected cross-file contradictions on adherence [12], and Arbiter catalogs interference in system prompts [13]. Applicability and precedence reasoning are therefore not new. Our contribution is narrower: a task and a traceable set of mined AGENTS.md pairs with controlled variants for separating valid overrides and disjoint applicability from unresolved conflict, and a same-evidence comparison of a direct and a structured model; human reference labels are the next step.

III. Task and Instruction Profile

A *case* is one pair of verbatim instruction spans (with file path, line numbers, heading, and enclosing section), one concrete task that edits one target file, one pinned commit, and one loading profile. The label concerns this pair and task, not the overall quality of a repository.

Profile P1 (frozen; an experimental interpretation of [1], not a claim about every tool). Only AGENTS.md files participate. The binding loaded set is every AGENTS.md on the target's ancestor directory chain; files in unrelated branches are not loaded. A loaded instruction is in force unless its own wording limits it. When two loaded instructions are incompatible on the same decision, the more deeply nested file wins, and ancestor instructions about other decisions stay in force. Equal authority (same file or same depth) is not ranked by order of appearance; only an explicit statement can create priority. User prompts, other file formats, imports, symbolic links, multiple working roots, and multi-target tasks are excluded.

Labels and decision sequence. The handbook orders the judgment as follows: (1) if meaning or applicability cannot be established, *Insufficient evidence*; (2) if, before priority, either instruction is inactive (not loaded, or excluded by its own condition), *Disjoint applicability*; (3) if the active instructions can be satisfied together, *Compatible* (different decisions, refinements, duplicates, or a preference beside a requirement); (4) otherwise *Valid override* if P1 or an explicit statement picks a winner, else *Unresolved conflict*. Modality is preserved: "prefer X" does not forbid Y, and descriptions or examples are not rules unless worded as requirements.

IV. Corpus and Annotation

Sampling. The discovery frame was the `context_files.csv` table of a public configuration dataset [5] (Zenodo record 20090356): 237 repositories whose rows show a root and at least one nested AGENTS.md. In a seeded random order we retrieved the first 30 through the GitHub API at their default-branch commit on 21 October 2026, storing every AGENTS.md as inert JSON with its SHA-256. Repositories had to carry an open-source license, not be forks or archived, have no symlinked AGENTS.md, and have a root and a nested file with at least five non-empty lines (24 of 30). Nested files under template, example, vendored, or hidden directories were excluded as separate project roots.

Candidate pairs. Retrieval is separate from classification and deliberately challenge-oriented: instruction units (list items and paragraphs outside code fences that contain a directive verb or modal) are paired when they share a specific term (a tool, command, or convention word) or name different values from one family (for example `npm` and `pnpm`); the score favors different values, opposite polarity, conditions, and precedence words. Verbatim duplicates are excluded. Repositories needed at least four candidate pairs, two of them across files (12 qualified); the first two in frame order formed the calibration split and the next eight the evaluation split, so no repository crosses splits. Up to four pairs per repository were selected by score with a quota for ancestor–descendant and unrelated-branch pairs, yielding 27 evaluation originals (16 ancestor–descendant, 5 unrelated-branch, 6 same-file) and 7 calibration originals. Tasks are researcher-authored from fixed templates (for example "make a small fix in `<target>` and complete the checks this change requires before committing"), and targets are real files under the relevant directory. This selected sample cannot estimate how often conflicts occur.

Strata. Each original has a *context-changing variant* that keeps both instructions' wording and changes one documented property: the target path (16 cases, moving the target outside the nested file's directory) or the placement of one instruction (11 cases, relocating it to the root or to a nested file on the target's chain). A relocated instruction is appended under a copy of its nearest heading, so outer headings of its original file are lost; no human has yet checked whether this changes meaning. Each original also has an *order control* that only swaps the presentation order of the two instructions. Variants and controls are constructed material and are reported separately from originals.

Annotation protocol (prepared, not yet carried out). A frozen handbook (v1.0) and a standalone offline tool support two independent annotators and an adjudicator: each annotator receives a separately randomized packet with opaque item IDs, without predictions, strata, or transformation notes, and sees the task, target, profile, the AGENTS.md paths with the target's directory chain marked, both spans with their sections and full files, and keyword-scanned scope statements. A 20-item calibration packet (six authored practice items with feedback) precedes the evaluation packet. These materials are released, but no human labels were collected for this submission, so RQ1 is not answered here and no agreement statistic is reported.

V. Methods and Setup

B0 applies Ruleherd's contradiction rules (v0.1.3, commit 94559fc6; keyword pairs such as npm/pnpm and tabs/spaces, and never/always polarity across different files) to the two spans of a case; it yields a binary flag and is scored only on conflict detection. **B1** is a direct contextual model: one call receives the evidence packet (task, target, commit, all AGENTS.md paths, both spans with line-numbered sections, and the scope statements), the full profile, the label definitions, and the decision sequence, but neither the full files nor the chain marking that the annotation tool shows. **B2**, the CIPHER pipeline, uses the same model and packet. One call extracts, for

each instruction, its requirement, regulated decision, modality, explicit condition, and whether that condition covers the task, plus any explicit priority statement. Code then computes the loaded set and depth precedence deterministically; an unclear condition is treated as covering the task, and B2 cannot abstain. A second call, made only when both instructions remain active and binding, judges whether they constrain the same decision incompatibly; fixed rules map the results to a label. Because B2 may use two calls, **B1R** adds an equal-call control: B1 followed by one self-review call over the same evidence.

All model arms used the open-weight Qwen3.6-35B-A3B [14] (35B parameters, 3B active; Q4_K_M quantization, Ollama digest a7eb95c5; served by Ollama 0.40.6, auto-updated to 0.40.7 before the last four B2 cases and the B1R run) on one laptop GPU with thinking enabled and the model card's recommended settings for precise coding tasks (temperature 0.6, top-p 0.95, top-k 20), a fixed seed, JSON-schema-constrained output, and at most two retries with a changed seed. Each request contained only the method prompt and the evidence, with no case identifier, stratum, or label. Calibration runs led to one change before any held-out run: B2's evidence-sufficiency field became an explicit two-value enum (B1's prompt was unchanged). Before any held-out output was inspected, an independent code review found that the header of the scope-statement block implied all listed files were loaded; it was corrected for every arm, all held-out runs were repeated, and the superseded runs were never opened. The profile, handbook, case manifest, prompts, schemas, and retry policy were frozen by hash before the reported held-out runs, which stayed sealed until the analysis below was frozen. The planned primary outcome, five-class macro-F1 against human labels, could not be computed; a plan addendum written before the predictions were opened defines the reported analysis.

VI. Results

Reference without human labels. Human labeling was not carried out for this submission, and no model output substitutes for it. Accuracy is therefore reported only where P1 alone fixes the label: 26 cases (5 originals, 16 target-path variants, 5 order controls) in which one instruction's file is not on the target's directory chain, so that instruction is not loaded and the label is *Disjoint applicability*. B2 computes the loaded set in code, so on these cases it is correct by construction unless extraction fails; the open question is whether B1 and B0 also get them right. All other measures in Table I need no reference. This analysis was defined and frozen by hash before the sealed predictions were opened.

Directory scope (RQ2, determined subset). B1, B2, and B1R each labeled all 21 profile-determined originals and variants correctly (0 cases favored B2 and 0 favored B1), so the direct model inferred directory applicability from the paths and the profile text without a deterministic step. B0, which ignores scope, flagged 2 of the 26 cases as conflicts: one ESM/CommonJS keyword match between two unrelated

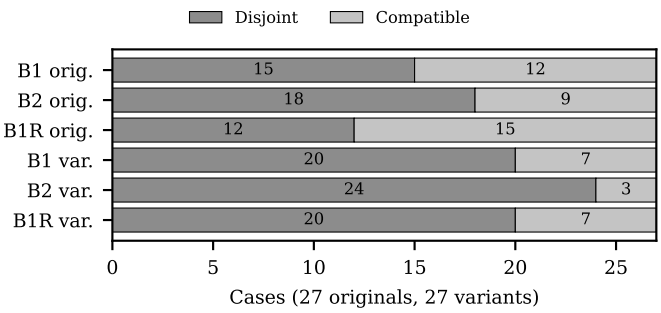


Fig. 1. Predicted labels per model arm on the 27 originals and 27 variants (counts; not accuracy). No arm predicted *Unresolved conflict*, *Valid override*, or *Insufficient evidence*.

packages, in the original and in its order control. For this model, the determined subset does not support the hypothesis that explicit structure is needed to handle directory scope.

Label-free comparison. No model arm labeled any original or variant an unresolved conflict or a valid override; every prediction was *Compatible* or *Disjoint* (Fig. 1), while B0 flagged 2 of the 54. The challenge-oriented retrieval thus surfaced lexically similar pairs that the models judged non-conflicting. The arms differ on the boundary between compatibility and condition-limited applicability: on originals B2 chose *Disjoint* 18 times and B1 15 times; B1 and B2 agreed on 22 of 27 originals (Cohen's $\kappa = 0.62$) and 23 of 27 variants ($\kappa = 0.53$). The first non-determined original on which they disagree (by the declared selection rule) pairs a root rule "use `Protocol` for decoupling" with a nested rule "avoid `TYPE_CHECKING`", both Python-specific, for a TypeScript target. B1 treated both rules as inactive; B2's extraction found no explicit condition and returned *Compatible*. Which reading is right is a judgment about implicit applicability that P1 does not settle and that human labels must resolve.

Stability and sensitivity (RQ3). Swapping only the order of the two instructions changed the label in four of 27 controls for each model arm (23, 23, and 23 unchanged for B1, B2, and B1R), although all 5 determined controls stayed correct. All 16 target-path variants, which move the target out of the nested file's directory, were labeled *Disjoint* by every arm; the label differed from the original's in 8 (B1), 7 (B2), and 11 (B1R) cases, because many originals were already judged inactive by condition. Placement variants changed B2's label 1 time and B1's 5 times out of 11; without reference labels we cannot say which changes are correct.

Cost. No arm had a failure, retry, or malformed output. B2 averaged 1.26 calls and 3589 output tokens per case, against 1.00 and 2177 for B1; the equal-call control B1R (3960 tokens) changed its first answer in 1 of 81 cases. Latency is not compared because the GPU was shared with another process for part of the runs.

VII. Threats to Validity

Without human labels, accuracy is measured only on the profile-determined subset, which B2 solves by construction

TABLE I

HELD-OUT RESULTS WITHOUT HUMAN LABELS. PD: PROFILE-DETERMINED CASES, WHOSE CORRECT LABEL UNDER P1 IS *Disjoint applicability* BECAUSE AN INSTRUCTION FILE IS OFF THE TARGET'S CHAIN (B2 COMPUTES THIS IN CODE). OTHER COLUMNS NEED NO REFERENCE. COST: MEAN CALLS / MEAN OUTPUT TOKENS PER CASE.

	PD correct	False conflicts	Called conflict	Order controls	Prediction changed in variants		Cost
Method	(orig.+var.)	on PD (all)	(orig.+var.)	same prediction	target path	placement	calls / tokens
B0	–	2/26	2/54	–	–	–	0.0 / 0
B1	21/21	0/26	0/54	23/27	8/16	5/11	1.0 / 2177
B2	21/21	0/26	0/54	23/27	7/16	1/11	1.3 / 3589
B1R	21/21	0/26	0/54	23/27	11/16	3/11	2.0 / 3960

and which proved easy for the direct model; the semantic question of condition-based and implicit applicability remains unmeasured, and agreement between methods is not accuracy. The sample is small (27 originals from eight repositories) and challenge-oriented, so it cannot estimate how often conflicts occur; the absence of predicted conflicts may reflect the retrieval as much as the repositories. Tasks were authored from templates, and variants and controls are constructed. One profile and one quantized open-weight model, run on a laptop GPU whose runtime received a patch update during the runs, limit generalization, and public repositories may appear in the model's training data. Results concern classification under P1, not whether agents follow instructions or produce better code.

VIII. CONCLUSION

CIPHER contributes a task definition, a traceable set of real AGENTS.md instruction pairs with controlled variants, and a blinded annotation protocol. In this preliminary evaluation without human labels, a local open-weight model given the full evidence resolved directory scope as reliably as a pipeline that computes it explicitly, and at lower cost, whereas a scope-blind keyword heuristic raised false alarms; none of the mined pairs was judged a genuine conflict. The open problem is condition-based and implicit applicability, where the methods disagree and an order swap alone alters some predictions. Collecting independent human labels for those cases, and testing other models and loading profiles, are the next steps.

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